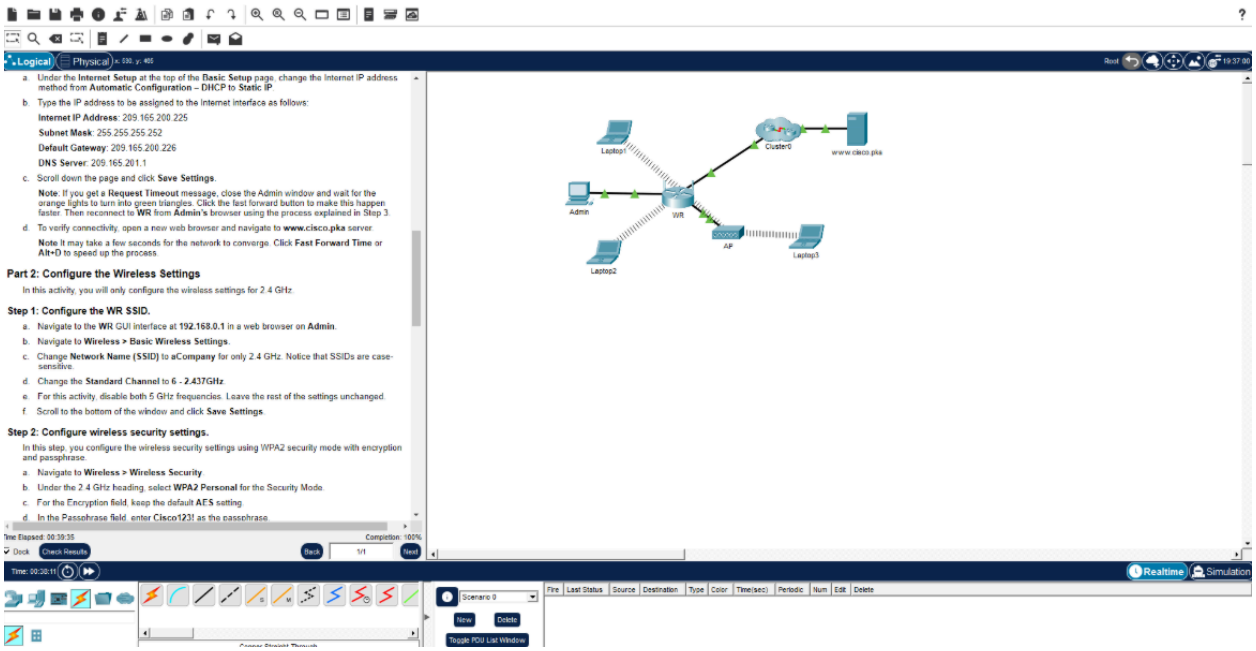
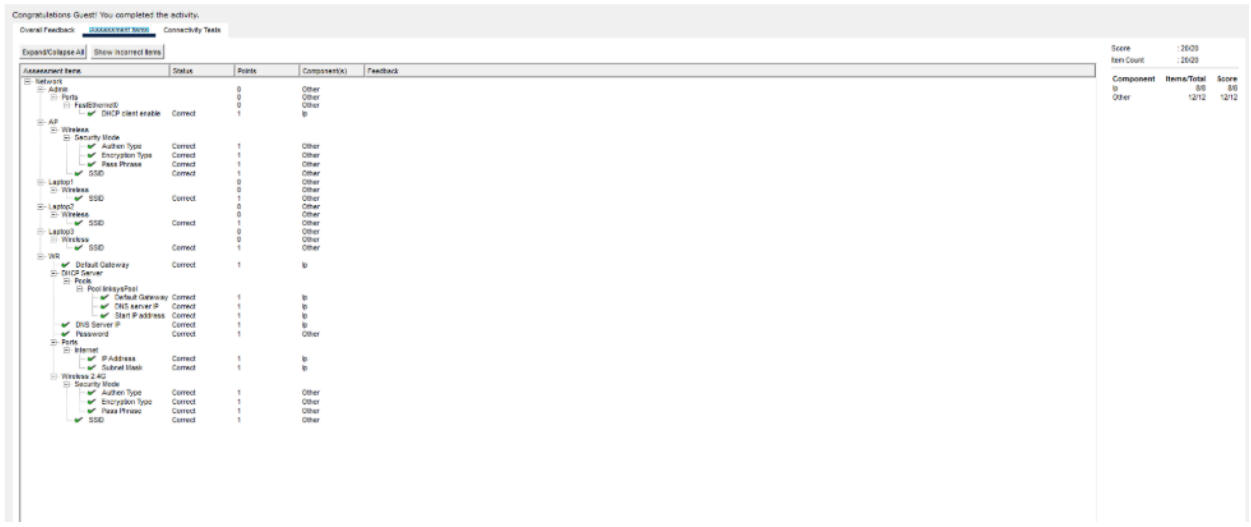
Lab 5: Packet Tracer - Configure a Wireless Network





Objectives

- Connect to a wireless router and configure its basic settings.
- Implement wireless security (WPA2 Personal) and SSID configurations.

- Connect both wired and wireless clients to the network.
- Extend wireless coverage using an Access Point (AP).
- Perform administrative tasks, including changing passwords and DHCP address ranges.

Topology Description

The network consists of a Wireless Router (**WR**) acting as the central gateway. A wired administration PC (**Admin**) is connected to the LAN for management. Two laptops (Laptop1 and Laptop2) connect via 2.4 GHz wireless. An Access Point (**AP**) is connected to the **WR** to extend the signal for **Laptop3**. The **WR** is connected to the internet via a static IP configuration to reach the **www.cisco.pka** server.

Addressing Table (Final Configuration)

Device	Interface	IP Address	Subnet Mask	Default Gateway
WR	Internet	209.165.200.225	255.255.255.252	209.165.200.226
WR	LAN (Loc...	192.168.50.1	255.255.255.0	N/A
Admin	NIC	192.168.50.100	255.255.255.0	192.168.50.1
Laptop1	Wireless	192.168.50.101	255.255.255.0	192.168.50.1
Laptop2	Wireless	192.168.50.102	255.255.255.0	192.168.50.1
Laptop3	Wireless	192.168.50.103	255.255.255.0	192.168.50.1

Router & Wireless Configurations

1. Wireless Router (WR) Settings

- **Internet Setup:** Static IP (209.165.200.225), DNS (209.165.201.1).
- **Wireless 2.4 GHz:** SSID: **aCompany**, Channel: **6**, 5GHz: **Disabled**.
- **Security:** WPA2 Personal, Encryption: **AES**, Passphrase: **Cisco123!**.
- **Administration:** Router Password changed to **cisco**.
- **LAN Setup:** Router IP **192.168.50.1**, DHCP Pool starting at **.100**, Max users **50**.

2. Access Point (AP) Settings

- **Port 1:** SSID: **aCompany**, Security: **WPA2-PSK**, Passphrase: **Cisco123!**.

Lab Questions and Answers

Part 1, Step 2: Configure Admin to use DHCP

- **What is the IP address of the computer?**
  192.168.0.100 (Initial DHCP assignment).
- **What is the subnet mask of the computer?**
  255.255.255.0.
- **What is the default gateway of the computer?**
  192.168.0.1.

Part 1, Step 3: Connect to the WR Web Interface

- **Is the IP address for Admin within this range? Is it expected? Explain your answer.**
  Yes, the IP address is within the range (typically 192.168.0.100 to 192.168.0.149). This is expected because the WR is acting as a DHCP server for the local network and assigns addresses from its pre-configured pool.

Part 4, Step 2: Change the DHCP address range in WR

- **The Web Browser will display a Request Timeout after a short time. Why?**
  The request timeout occurs because the router's IP address (the management

gateway) was changed from 192.168.0.1 to 192.168.50.1. The Admin PC is still on the 192.168.0.0/24 subnet, and the existing session is no longer valid on the new network segment.

- **What is the new IP address information for Admin?**
  192.168.50.100 (after performing an `ipconfig /renew`).
- **Notice that Laptop1 connected to the AP instead of WR. Why?**
  Wireless devices typically connect to the source with the strongest signal strength. Since the AP and WR share the same SSID, Laptop1 automatically roamed to the Access Point because it provided a better signal quality (RSSI) in its specific location compared to the Wireless Router.